AI for Medical Diagnosis (DeepLearning.AI, Coursera)

Completion Evidence — Jacob Lacy

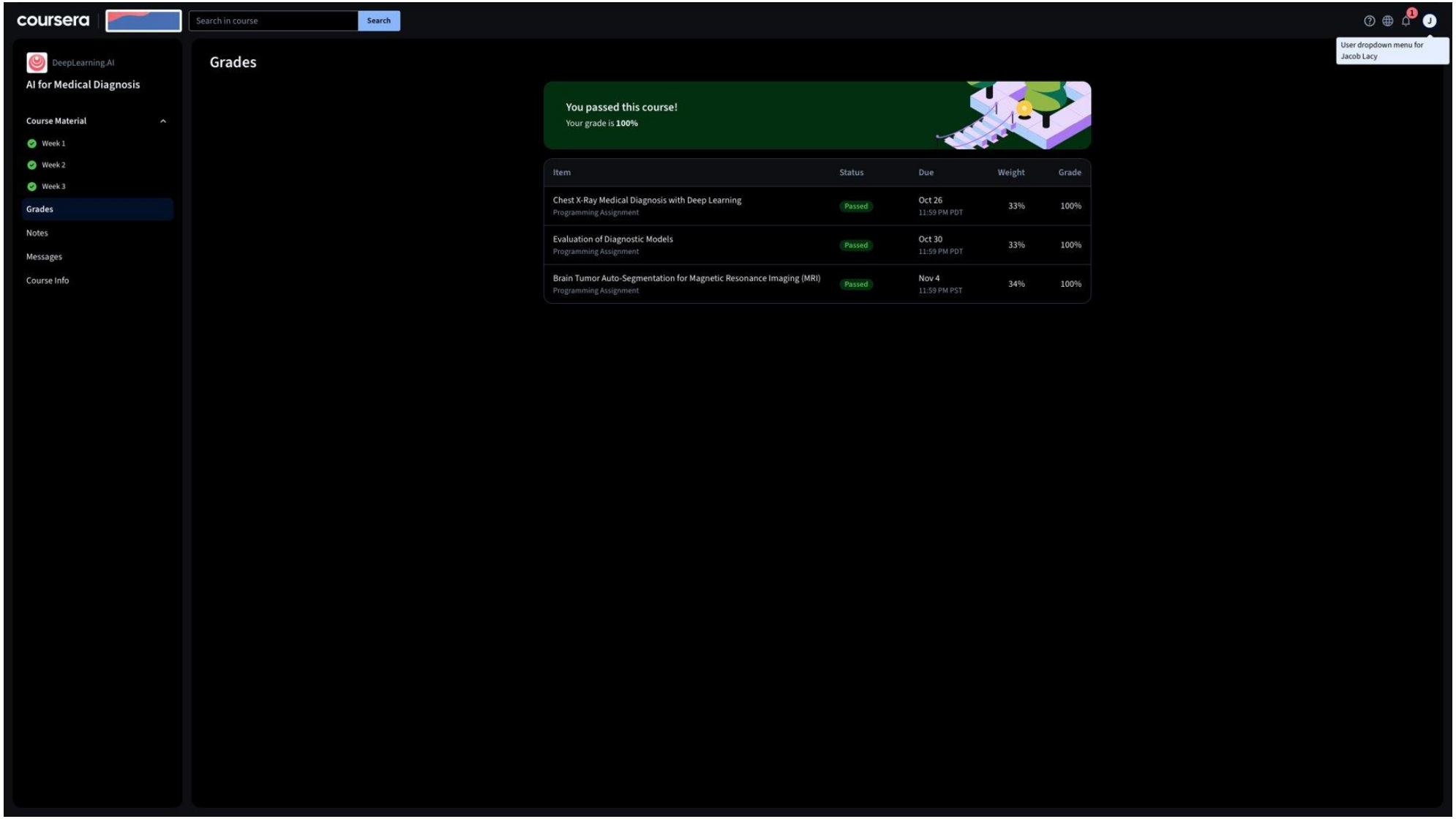
TRM 1B - Artefact A5

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Course completed with a final grade of 100%. Graded work was completed within Coursera's free trial and the paid certificate was not purchased, so completion is shown through the grades page, the completion email and the module completion pages. The three graded programming assignments are provided separately as Artefact A6.

Grades Page: Course Passed (100%)



Course Completion Email

Hi Jacob Lacy,

Congratulations! You have completed **AI for Medical Diagnosis** - course 1 of the AI for Medicine Specialization.

So what can you do now?

1. Continue on to AI for Medical Prognosis - course 2 in this Specialization.
2. Subscribe to The Batch, our weekly newsletter, for the latest developments in AI.
3. Follow us on Facebook | LinkedIn | Twitter to stay updated about course launches and events near you.

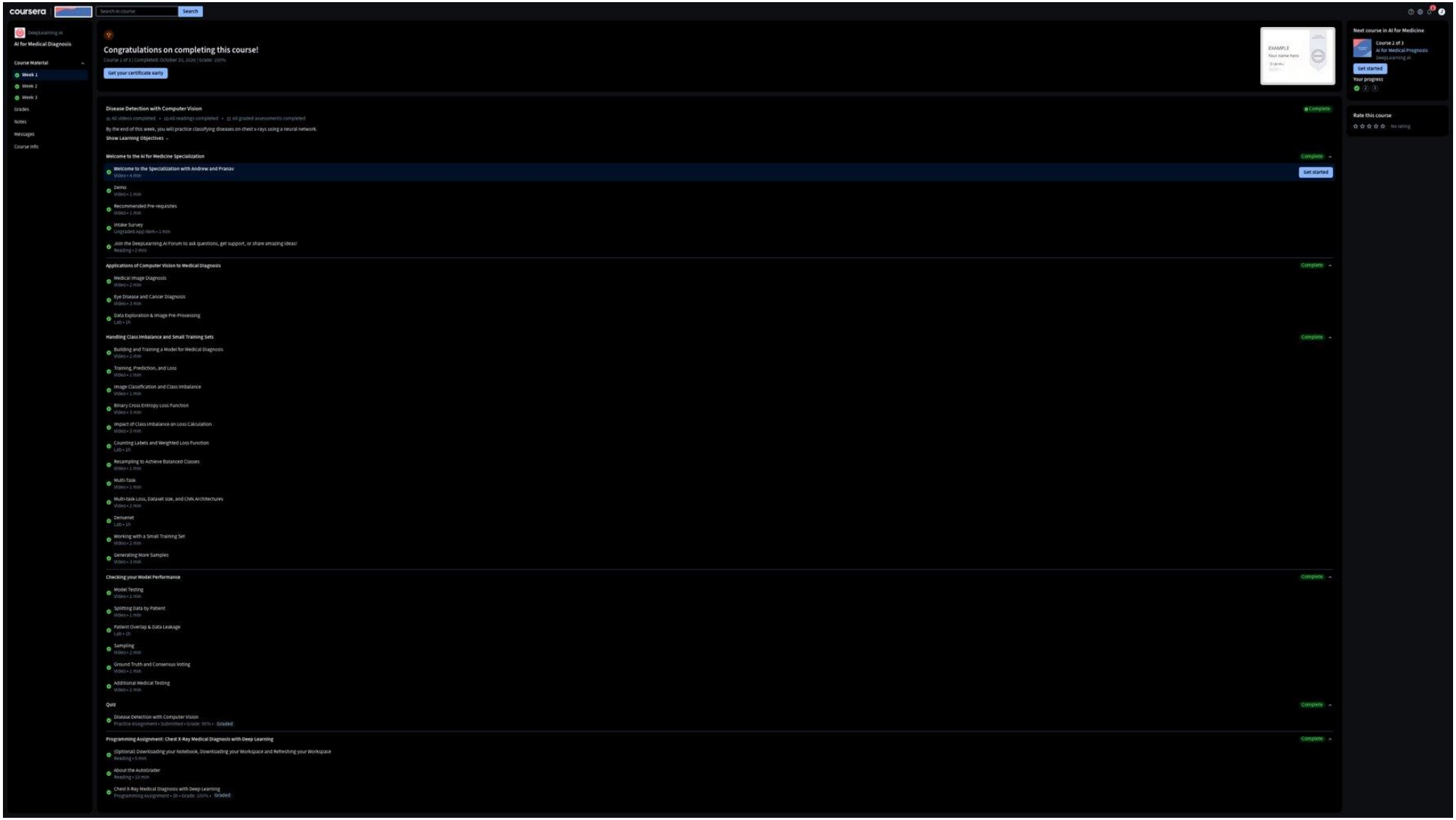
Keep learning!

The DeepLearning.AI team

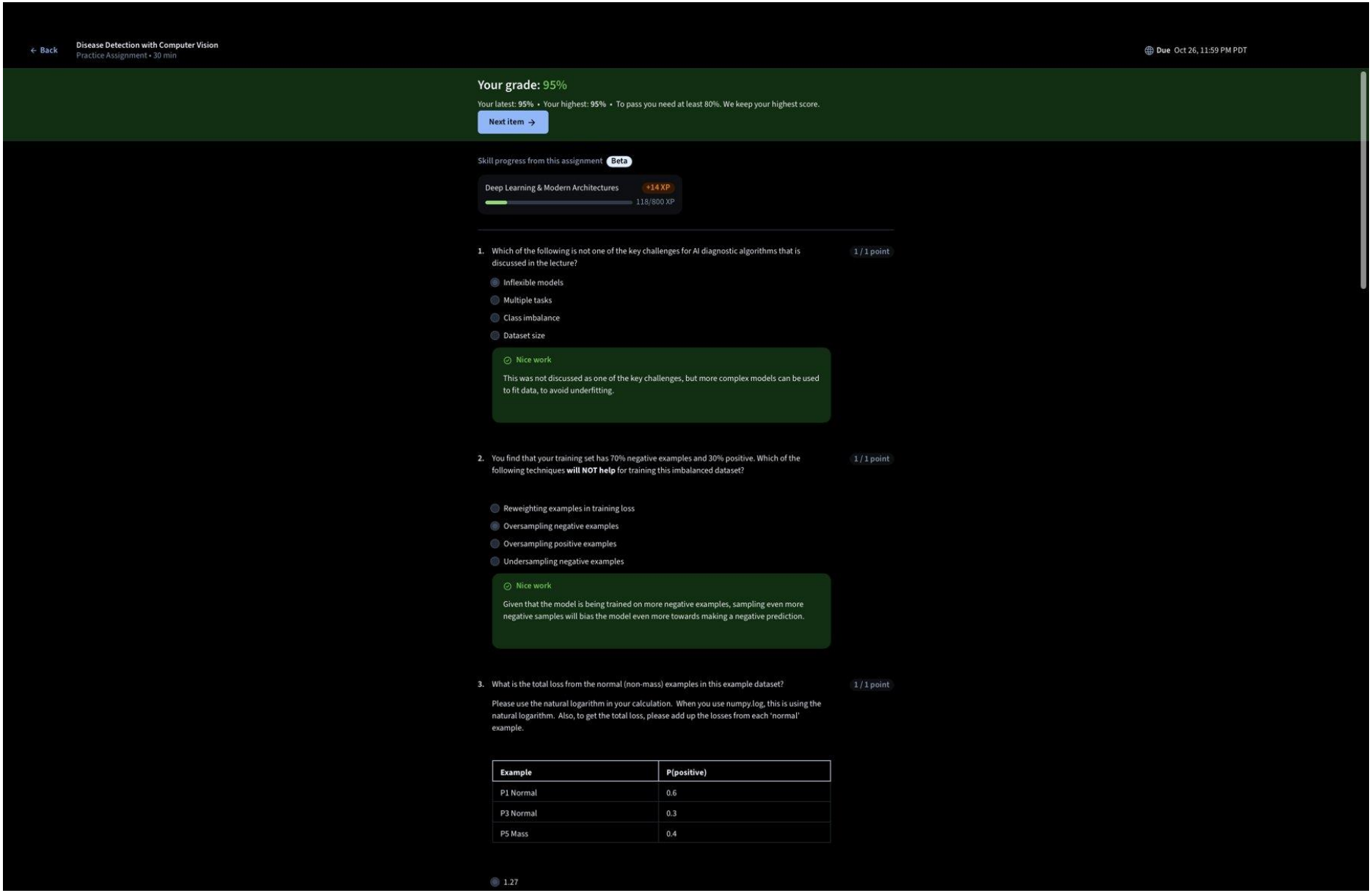
View completion

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Week 1 – Disease Detection with Computer Vision: Module Completion



Week 1 – Quiz Review (95%)



3. What is the total loss from the normal (non-mass) examples in this example dataset? 1 / 1 point

Please use the natural logarithm in your calculation. When you use numpy.log, this is using the natural logarithm. Also, to get the total loss, please add up the losses from each 'normal' example.

Example	P(positive)
P1 Normal	0.6
P3 Normal	0.3
P5 Mass	0.4

- (●) 1.27
- ( ) 2.19
- ( ) 0.00
- ( ) -0.4

Nice work

Since these are negative examples, the losses will be $-log(1 - P(positive))$.

For P1, $-log(1 - 0.6) = 0.91$.

For P3 $-log(1 - 0.3) = 0.36$.

The sum is $0.91 + 0.36 = 1.27$.

4. What is the typical size of medical image dataset? 1 / 1 point

- (●) ~10 thousand to 100 thousand images
- ( ) ~1 to 1 hundred images
- ( ) ~ 1 hundred to 1 thousand images
- ( ) ~1 million or more images

Nice work

Most often datasets will range from 10,000 to 100,000 labeled images. Fewer than 1000 is typically too few to train, validate and test a classifier, and very few datasets will have millions of images due to the cost of labeling.

5. Which of the following data augmentations would be best to apply? 1 / 1 point



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None of the above

Nice work

This rotation is most likely to help. This is a realistic transformation. Also, it does not risk changing the label.

6. Which of the following are valid methods for determining ground truth? Choose all that apply. 1 / 1 point

Confirmation by CT scan

Nice work

A CT scan can provide an objective ground truth. Keep in mind that there are likely fewer data examples where patients have both the chest x-ray and an additional diagnostic test for the same disease.

Consensus voting from a board of doctors

Nice work



6. Which of the following are valid methods for determining ground truth? Choose all that apply. 1 / 1 point

- [x] Confirmation by CT scan

  Nice work

  A CT scan can provide an objective ground truth. Keep in mind that there are likely fewer data examples where patients have both the chest x-ray and an additional diagnostic test for the same disease.

- [x] Consensus voting from a board of doctors

  Nice work

  Consensus is considered less reliable than biopsy verification. However, the limited availability of biopsy data means that consensus voting may still be the best (or only viable) option.

- [x] Biopsy

  Nice work

  Biopsy is definitely a valid method. Keep in mind that there are likely fewer data examples where patients have both the chest x-ray and an additional diagnostic test for the same disease.

7. In what order should the training, validation, and test sets be sampled? 1 / 1 point

- ( ) Validation, Test, Training
- (•) Test, Validation, Training
- ( ) Validation, Training, Test
- ( ) Training, Validation, Test

Nice work

First the test dataset should be sampled, then the validation set, then the training set. This is so that you can make sure you can adequately sample the test set, and then sample the validation set to match the distribution of labels in the test set.

8. Why is it bad to have the same patients in both training and test sets? 1 / 1 point

- ( ) Leaves too few images for the test set
- (•) Overly optimistic test performance
- ( ) None of these above
- ( ) Leaves too few images for the training set

Nice work

Having images from the same patient is bad because it has been shown that the model may learn patient-specific features that are not generalizable to other patients.



9. Let's say you have a relatively small training set (~5 thousand images). Which training strategy makes the most sense? 1 / 1 point

- [ ] Retraining all layers of a pre-trained model
- [ ] Retraining the first layer of a pre-trained model
- [x] Retraining the last layer of a pre-trained model

- Training, Validation, Test

Nice work

First the test dataset should be sampled, then the validation set, then the training set. This is so that you can make sure you can adequately sample the test set, and then sample the validation set to match the distribution of labels in the test set.

8. Why is it bad to have the same patients in both training and test sets? 1 / 1 point

- Leaves too few images for the test set
- Overly optimistic test performance
- None of these above
- Leaves too few images for the training set

Nice work

Having images from the same patient is bad because it has been shown that the model may learn patient-specific features that are not generalizable to other patients.

9. Let's say you have a relatively small training set (~5 thousand images). Which training strategy makes the most sense? 1 / 1 point

- [ ] Retraining all layers of a pre-trained model
- [ ] Retraining the first layer of a pre-trained model
- [x] Retraining the last layer of a pre-trained model

Nice work

By using a pre-trained model, you can make use of its ability to recognize lower level features, and then fine tune the last few layers using your dataset.

- [ ] Train a model with randomly initialized weights

10. Now let's say you have a very large dataset (~1 million images). Which training strategies will make the most sense? 0.5 / 1 point

- [ ] Retraining the last layer of a pretrained model
- [x] Training a model with randomly initialized weights.

Nice work

Given a very large dataset, you have the option of training a new model instead of using a pre-trained model.

- [x] Retraining the first layer of a pretrained model

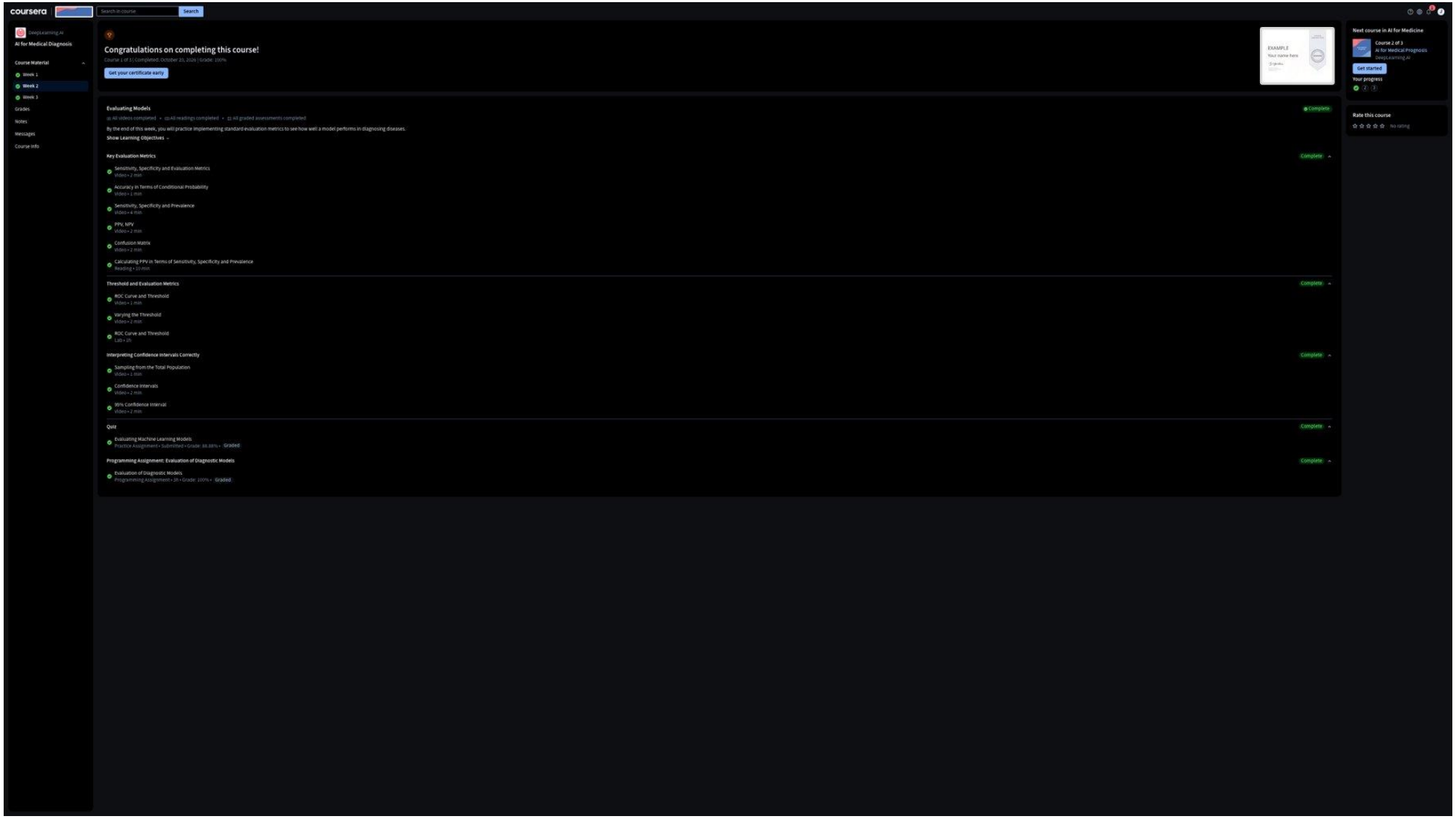
This should not be selected

Retraining the first layer of a pre-trained model would train the model on lower level features, but not necessarily help the model learn the higher level features of your dataset.

- [ ] Retraining all layers of a pretrained model



Week 2 – Evaluating Models: Module Completion



Week 3 – Image Segmentation on MRI Images: Module Completion

